

Fischer–Tropsch synthesis on mono- and bimetallic Co and Fe catalysts in fixed-bed and slurry reactors

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Abstract

Mono- and bimetallic cobalt and iron oxide nanoparticles deposited on the surface of a silica substrate were tested in a fixed-bed reactor and stirred-tank slurry reactor under different process conditions, for the Fischer–Tropsch reaction, this comparison is presented for the first time, being presented in this work a comparison of above systems in both reactor types. For the fixed-bed reaction, the monometallic iron-based catalyst presented the highest activity at low reaction temperatures. By contrast, the monometallic cobalt catalyst is the less active for all reaction temperatures. This catalyst records high values for the α parameter, whereas the iron catalyst reveals major selectivity to alcohols. Moreover, the introduction of iron produces a decrease in hydrocarbon chain growth and an increase in alcohol selectivity. This behaviour might be due to the formation of a Co–Fe alloy, observed by several characterization techniques. In the slurry studies, CO conversion is greater for the cobalt-based catalyst and lower for bimetallic systems, whilst the iron catalyst deactivates in a few hours, which is explained by the formation of carbonaceous deposits. The bimetallic catalysts record a decrease in CO conversion and an enhancement in the selectivity of methane and light hydrocarbons with the increase in iron amount. The absence of alcohols in this type of reactor is explained by the greater solubility of hydrogen, as well as by the greater contact time of reactants, which decreases the possibility of non-dissociative CO insertion.

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1. Introduction

The conversion of syngas (CO + H₂ mixtures) into liquids, and more specifically clean fuels and chemical feedstock via Fischer–Tropsch synthesis (FTS), is currently of increasing interest. This catalytic synthesis leads to a wide variety of products, whose abundance depends on the catalysts employed, as well as on operating conditions [1,2]. Although all VIII group metals [3] display some activity in the C–C coupling reaction during CO hydrogenation, the most active metals for FTS are ruthenium, nickel, iron and cobalt [4–7], with iron or cobalt-based catalysts being the most commonly used at industrial scale.

Cobalt catalysts record higher yields and long lifetimes, producing predominantly linear alkanes and heavy hydrocarbons, with low water gas shift activity [8]. By contrast, iron catalysts have a high water gas shift activity, high selectivity to both olefins and oxygenated products and appear to be stable when synthesis gas with a high H₂/CO ratio is converted [1,9]. In order to consider the benefits of both metallic catalysts, certain studies of bimetallic systems have been reported [10–14]. These studies indicated a catalytic behaviour in FTS that is different to the sum of both isolated metals. The accepted mechanism reaction is based on a pseudo-polymerisation of –CyH_x– units, which are adsorbed on the catalyst surface. The products are obtained according to a distribution known as Anderson–Schulz–Flory (ASF) [15]. This model establishes a linear relation between product yield (molar) logarithm and carbon atom number.

Two types of reactors are widely employed in FTS in industry: the multi-tubular fixed-bed reactor and the slurry bubble column reactor. Sie and Krishna [16,17] compared the advantages and disadvantages of both types of reactors.

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The major drawbacks of the slurry reactor are the need for continuous separation between catalyst and liquid products, a smaller scaling-up factor in comparison to the multi-tubular reactor and possible attrition of catalyst particles. The advantages are [18]: (i) low pressure drop in the reactor; (ii) excellent heat transfer characteristics resulting in stable reactor temperatures; (iii) no diffusion limitations; and (iv) possibility of continuous refreshment of catalyst particles.

Then the aim of the present work is to analyse the effect of incorporating both cobalt and iron species supported over a silica matrix in the FTS to produce liquid hydrocarbons, and the possible synergic effect of the addition of both metals. In addition, the influence of both fixed-bed and slurry reactors has been studied on the activity and selectivity of mono- and bimetallic cobalt and iron systems, as well as the influence of operation conditions. Studies that, as far as we know, after a deep revision of open literature, have not been published in previous works. Catalytic performance will be related to a wide characterization study of the Co/Fe catalysts used, which has been described in a previous paper [19] in which several characterization techniques have been used.

2. Experimental

2.1. Catalyst preparation

Silica-supported cobalt and/or iron catalysts were prepared by the wetness impregnation method on a silica carrier (Grace Davison, 310 m²/g, 1.22 m²/g) with cobalt nitrate (Co(NO₃)₂·6H₂O and Fe(NO₃)₃·9H₂O aqueous solutions. Cobalt and iron monometallic catalysts were prepared with appropriate amounts of the above salts to achieve a final metal amount of 10% (Co10-c and Fe10-c). Two bimetallic FeCo-supported catalysts were prepared containing a fixed amount of cobalt (10%) and different amounts of Fe (1 and 5%), CoFe10/1-c and CoFe10/5-c. Cobalt and iron salts were incorporated simultaneously. All impregnates were dried at 393 K overnight and calcined at 773 K under an airflow for 2 h.

2.2. Characterization techniques

The powder X-ray diffraction (XRD) patterns of the precursor and calcined samples were recorded using a Seifert 3000 P diffractometer, using nickel-filtered Cu K α 1 (λ = 0.15406 nm) radiation. A scanning step of 0.02° was taken between 5 and 80° Bragg' angles. Transmission electron micrographs were taken with a Fei Tecnai G30 microscope. The acceleration voltage was set at 200 kV. Surface analysis was carried out on a VG Escalab 200R electron spectrometer provided with an Mg K α X-ray source and a hemispherical electron analyser. The binding energies were referenced to the C1s peak at 284.9 eV due to adventitious carbon. Mössbauer spectra were recorded in a sinusoidal mode using a transmission spectrometer with a ⁵⁷Co/Rh source.

2.3. Activity measurements

Activity tests were carried out in both liquid phase slurry and fixed-bed catalytic reactors. In both cases, catalysts were activated before reaction by a reduction treatment with 100 ml_N/min, H₂ (20%)/N₂ (80%) for 2 h at 773 K. This activation procedure was carried into to account the TPR obtained results published in an earlier article [19].

The fixed-bed reactor consists of a 316 stainless steel tubular reactor (1 in. diameter), in which 0.3 g of catalyst (0.25–0.40 mm) were placed. The gas flows (He, H₂ and CO) were adjusted by mass flow controllers (Bronkhorst Hi-Tech), with an H₂:CO ratio of 2:1 (total flow = 100 ml_N/min, pressure = 20 bar). Reaction pressure was adjusted by a needle valve. The entire reactor system was placed inside a “hot box” with all lines heated (523 K) to avoid condensations. Reaction products were analyzed online by gas chromatography (Hewlett-Packard 5890), equipped with two semicapillary columns (SPB-5 and PoraPlot Q) and TCD (thermal conductivity detector) and FID (flame ionization detector). Conversion and selectivity results have been obtained after 16 h of reaction once steady state has been reached.

The liquid phase slurry reactor consists of a stainless steel vessel with a volume of 100 cm³ (Autoclave Engineers, mod. Eze Seal). The reactor was loaded with 0.5 g of catalyst, and *n*-hexadecane was used as solvent (25 ml). A gas mixture of H₂/CO (2:1) was passed through the reactor with continuous flows adjusted by mass flow controllers (Bronkhorst Hi-Tech). The pressure of the system was kept at the desired value with a pressure controller (Bronkhorst Hi-Tech). The liquid reaction product was analyzed by a Hewlett Packard 5890 gas chromatograph equipped with HP-1 column, which allowed for the separation of alcohols and remaining hydrocarbons; gas reaction products were analyzed by an Agilent 6890 gas chromatograph equipped with two columns in series: a packed Hayesep D and a HP-plotQ. The reaction conditions were studied in a previous paper with a silica-supported cobalt catalyst using this type of reactor [5], with the optimum reaction conditions, 4.0 MPa and 503 K, being those used here. Conversion, selectivity and olefinitiy are defined as follows:

$$\text{Conversion}(\%) = \frac{\text{mol CO total consumed}}{\text{mol CO in the feed}} \times 100$$

$$\text{Selectivity}(\%) = \frac{\text{mol CO to product}}{\text{mol CO total consumed}} \times 100$$

$$\begin{aligned} \text{Olefinitiy}(\%) \\ = \frac{\text{mol alkenes formed}}{(\text{mol alkanes formed} + \text{mol alkenes formed})} \times 100 \end{aligned}$$

3. Results and discussion

3.1. Characterization overview

The characterization of these systems has been reported in a previous paper [19]. The results provided by X-ray diffraction,

Table 1
Characterization properties of calcined and reduced catalysts

Sample	DRX phases	Crystal size (nm)	Mössbauer phases	%
Co10-c	Co ₃ O ₄	21	–	–
Fe10-c	Fe ₂ O ₃	18	Fe ₂ O ₃ Fe ₂ O ₃ (nano-size)	71 29
CoFe10/5-c	Co ₃ O ₄ Fe ₂ O ₃	14 14	– Fe ₂ O ₃ Fe ₂ O ₃ (nano-size)	– 69 31
CoFe10/1-c	Co ₃ O ₄ Fe ₂ O ₃	17 –	– –	– –
Co10-c (H ₂)	Co ⁰	17	–	–
Fe10-c (H ₂)	Fe ⁰	22	Fe ⁰ Fe ₂ O ₃ (nano-size) Fe _{1-x} O Fe ²⁺ (Fe ₂ SiO ₄)	39 20 38 3
CoFe10/5-c (H ₂)	Co ⁰ Co–Fe	10 23	Co–Fe Fe ₂ O ₃ (nano-size)	76 24
CoFe10/1-c (H ₂)	Co ⁰ Co–Fe	10 11	– –	– –

Mössbauer and X-ray photoelectron spectroscopy revealed the formation of separate phases of Co₃O₄ and Fe₂O₃ in bimetallic calcined samples, but in no case were cobalt–iron mixed oxides detected. Results from XRD patterns, transmission electron micrographs, photoelectron and Mössbauer spectra of hydrogen-reduced samples were conclusive on the formation of both metallic Co⁰ and Fe⁰ in monometallic samples, whilst bimetallic phases comprise mainly two phases: Co⁰ and an iron–cobalt alloy. It was also found that the crystallite size of the alloyed phase increased with iron content, whereas that of Co⁰ remained almost constant. A summary of some of the results obtained is presented in Table 1.

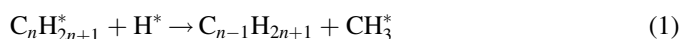
3.2. Fixed-bed reactor

The activity and selectivity shown by mono- and bimetallic cobalt and iron catalysts for FTS using a fixed-bed reactor will be now be addressed in depth, although preliminary results have been described in a previous paper [11]. The influence of temperature and type of active phase in CO conversion and product distribution will be analyzed.

Fig. 1 shows the variation of CO conversion versus temperature for all the catalysts used. In all cases, conversion increases with reaction temperature. The monometallic iron sample (Fe10-c) records higher conversion values in the entire reaction temperature range studied, except at 533 K, where the CoFe10/5-c catalyst presents the highest conversion, of around 60%. In contrast, its cobalt counterpart (Co10-c) exhibits the lowest conversion values. This result is related to the low dispersion of cobalt phase on the silica substrate. Previous studies in fixed-bed reactor showed that metal dispersion has a major influence on activity and selectivity, decreasing conversion with dispersion decrease [20]. The surface

characterization results of this catalyst, obtained by XPS and widely described in a previous paper [19], showed low dispersion of cobalt metal particles, which could explain the low CO conversion values. The addition of iron to the cobalt catalyst (CoFe10/1-c and CoFe10/5-c) led to an increase in CO conversion in relation to Co10-c; with the CoFe10/5-c catalyst being the most active at 533 K. This effect is very pronounced for the CoFe10/1-c sample, as a small amount of iron leads to a dramatic increase in conversion in relation to the monometallic cobalt catalyst at 533 K (Fig. 1). This promotion effect was attributed to the formation of a cobalt–iron alloy [11] that produces a reduction in cobalt particle size through the formation of this Co–Fe alloy, and therefore an increase in active metal phase dispersion [19].

Product selectivity distribution at different reaction temperatures is displayed in Fig. 1. The results obtained show that active sites favour methane formation more than chain growth, since the main product for the whole catalyst is methane. This high yield to CH₄ can be explained in terms of FTS mechanisms reported in the literature [21], finding that the probability of chain termination for methane is between 5 and 20% higher than for paraffins. A further fact to be considered for explaining the high yield towards methane is the heat and mass transfer limitation produced in this type of reactor. Mass transfer limitations lead to an increase in thermodynamically favoured products (methane). Moreover, the existence of hot points due to the high reaction heat produces a decrease in the chain growth parameter and an increase in the yield toward methane. In addition, there may be secondary reactions by metal groups, such as hydrogenolysis [22,23]



In contrast, the selectivity to C₂ hydrocarbons is low. Wojciechowski et al. [21] explained this decrease in C₂ selectivity by the mobility and reactivity of these C₂ groups, and their facility to bind on metallic sites, where they react quickly (hydrogenation, hydrogenolysis and chain growth).

These observations are reflected in the hydrocarbon distribution (Fig. 2). At 513 K, the Co10-c catalyst recorded

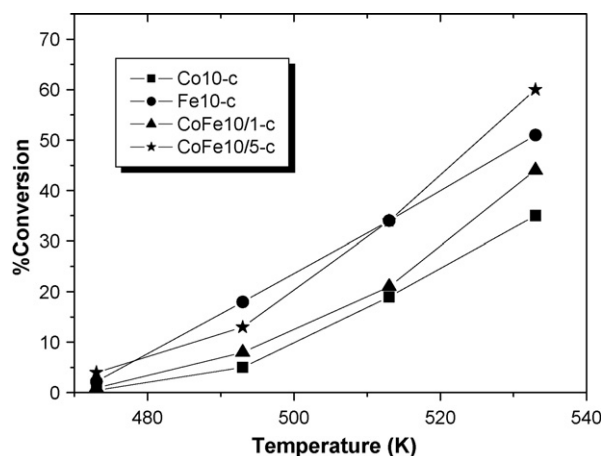


Fig. 1. CO conversion for mono- and bimetallic catalysts in a fixed-bed reactor for different temperatures ($P = 20$ bar, $H_2/CO = 2:1$).

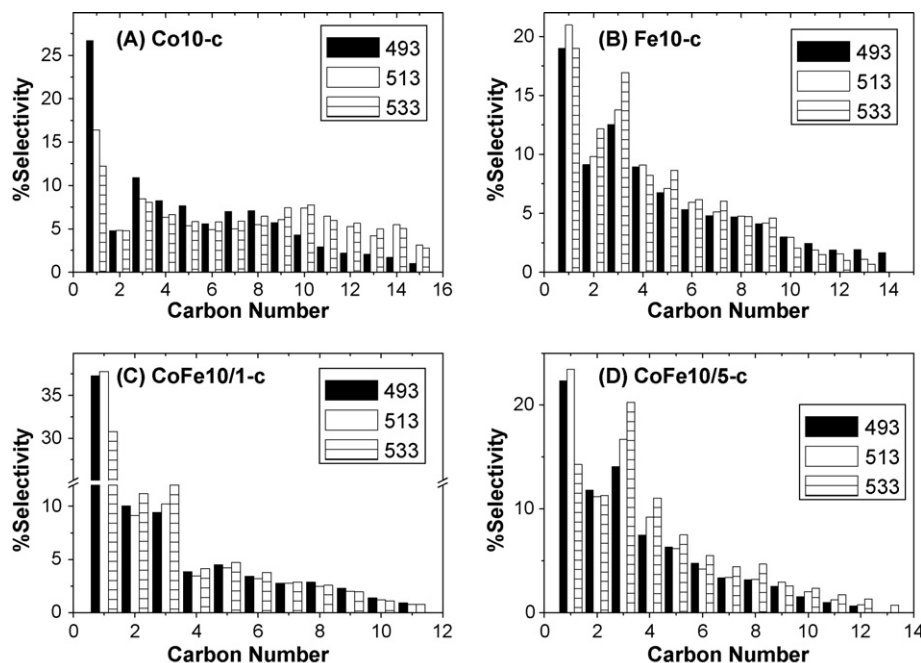


Fig. 2. Selectivity towards hydrocarbons for mono- and bimetallic catalysts in a fixed-bed reactor for different temperatures ($P = 20$ bar, $H_2/CO = 2:1$).

an average selectivity value of around 7% for all hydrocarbons between C_5 – C_{12} , which corresponds to the gasoline and diesel fractions. At 533 K, an increase is observed in the selectivity to higher hydrocarbons (C_{10} – C_{16}). This effect is accompanied by an increase in CO conversion. Catalyst Fe10-c, by contrast, exhibited higher selectivity to lighter hydrocarbons [24] (Fig. 2). By the other hand, the distribution of high hydrocarbons for CoFe10/5-c is very similar to that obtained for Fe10-c. While for CoFe10/1-c, the selectivity toward high hydrocarbons is lower due to the elevated production of C_1 . In both bimetallic systems, the major products are light hydrocarbons (C_1 – C_4). However, a selectivity increase toward longer chain hydrocarbons (from C_3) occurred with the increase in reaction temperature. This behaviour is accompanied by a diminution in methane formation at 533 K and an increase in the probability of chain polymerization.

The chain growth probability parameter (α) was calculated for all systems using the Anderson–Schulz–Flory (ASF) procedure [15] (Table 2), calculated from C_4 to C_{14} products. Co10-c shows higher α values for all the reaction temperatures studied, indicating the formation of long chain hydrocarbons in the gasoline and diesel range, with a sharp increase from 493 to

513 K. The lower α values observed for the iron monometallic sample indicate a lesser formation of long chain hydrocarbons. The growth probability parameter decreases with reaction temperature due to the formation of light products, with this behaviour being typical of iron systems. The results obtained in both bimetallic systems show that the inclusion of iron in the cobalt catalyst leads to a decrease in chain growth, with the α parameter being similar to the iron monometallic catalyst and almost constant for 513 and 533 K.

Alcohol product distribution is quite different (Fig. 3). Co10-c catalyst showed a very low selectivity to alcohols ($<1.5\%$), as is expected for cobalt catalysts [25], with ethanol being the main alcohol product for all temperatures. These results confirm that the cobalt catalyst is not a good candidate for producing alcohols, since the CO insertion rate in CH_x fragments is lower than the CO dissociation and polymerization reactions. The Fe10-c sample presents slightly higher selectivity to alcohols than the cobalt catalyst, with methanol being the main alcohol product at low temperatures, switching to C_3OH at higher temperatures (Fig. 3). However, overall alcohol selectivity decreases with temperature because non-oxygenated hydrocarbons are thermodynamically more stable at higher temperatures.

Several factors can be argued to explain the behaviour of cobalt and iron monometallic catalysts: (i) the cobalt catalyst possesses a major capacity for absorbing CO dissociatively and only ethanol formation by CO insertion in C_1 adsorbed fragment is produced with any reasonable probability; (ii) the dissociation of CO is more difficult on the iron-based catalyst, which is a good alcohol producer; (iii) methanol formation, by the hydrogenation of non-dissociative CO, is a reaction favoured at low temperature, with methanol being the major alcohol product.

Table 2

ASF chain growth parameter for all systems in a fixed-bed reactor for different reaction temperatures ($P = 20$ bar, $H_2/CO = 2:1$)

Temperature	α^a			
	Co	Fe	CoFe10/1-c	CoFe10/5-c
493	0.75	0.74	0.72	0.66
513	0.86	0.70	0.70	0.68
533	0.86	0.66	0.68	0.68

^a Calculated for C_4 – C_{14} .

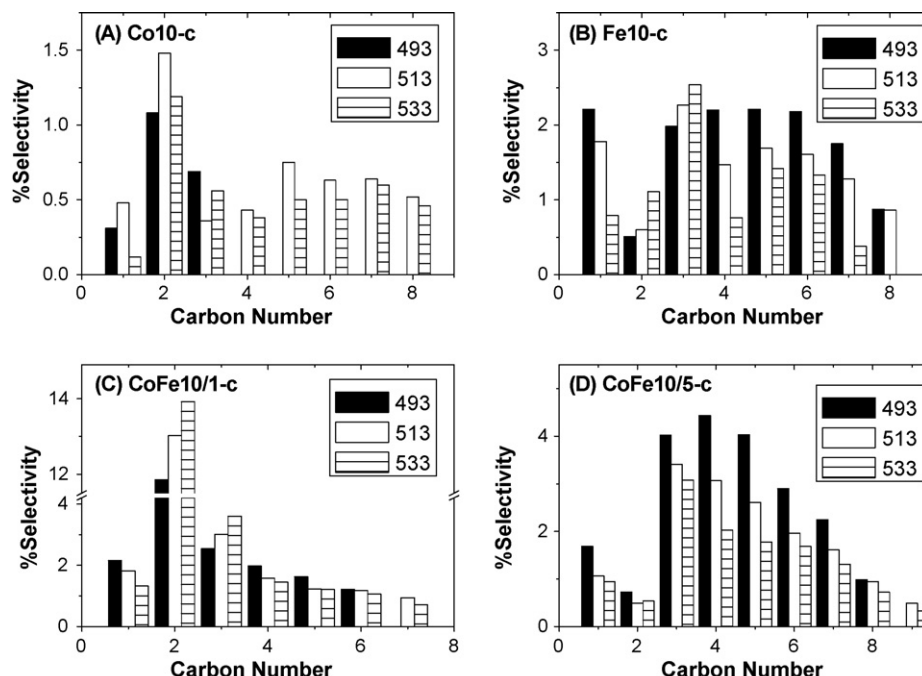


Fig. 3. Selectivity towards alcohols for mono- and bimetallic catalysts in a fixed-bed reactor for different temperatures.

The introduction of 1% iron into the cobalt catalyst (CoFe10/1-c) produced a dramatic increase in the selectivity to ethanol, and this effect was very evident at 533 K. However, the increase in the amount of iron from 1 to 5% changes the oxygenated distribution products, with an appreciable decrease in the selectivity to ethanol, and a selectivity increase toward longer chain alcohols (C_3 – C_6). The behaviour of bimetallic catalysts regarding increased selectivity toward alcohols is attributed to the presence of a Co–Fe alloy [11], and depends on the Co/Fe ratio. A small amount of Fe produced an alcohol distribution similar to that shown by the monometallic Co10-c catalyst, but one order of magnitude higher, thereby causing high C_2 OH selectivity. However, CoFe10/5-c showed a product distribution similar to the iron catalyst (Fe10-c), but once again producing more alcohols with the bimetallic system.

In the case of bimetallic catalysts, the appearance of this distribution particular to oxygenated products might be explained by the alkyl mechanism, since this explains the formation of both hydrocarbons and oxygenated products [26]. Thus, for CoFe10/1-c catalyst, the major products formed are CH_4 and ethanol. The results obtained for CoFe10/1-c catalyst indicate that the addition of a low amount of iron to the monometallic cobalt system gives rise to CO hydrogenation to produce methane and non-dissociative insertion of CO in a surface alkyl species to produce oxygenated products. Thus, the formation of a small proportion of a Co–Fe alloy increases chain termination, producing methane by hydrogenation of methylene, or ethanol by the insertion of CO in a methylene group.

The behaviour of CoFe10/5-c is similar to Fe10-c, decreasing selectivity to methane and increasing the formation of C_3 hydrocarbons and alcohols in relation to CoFe10/1-c. The higher amount of iron in CoFe10/5-c increases alloy formation

and decreases the size of cobalt and iron particles, thereby increasing the dispersion of active phases [19]. This should favour the development of the chain growth reaction through the higher proportion of surface cobalt, thus resulting in a lower selectivity to methane and to longer alcohol chain formation.

Considering these results, it can be concluded that alloy formation in bimetallic catalysts enhances the ability for CO insertion reaction, thus increasing alcohol formation, and therefore decreasing the possibility of direct chain growth.

3.3. Liquid phase slurry reactor

A previous study on the behaviour of the cobalt catalyst in FT reaction using a slurry reactor revealed an increase of CO conversion and a decrease in deactivation with pressure increase [5], consistent with that reported in literature [27]. The higher pressure-induced reaction rate is explained by the increase in the solubility of both CO and H_2 in the solvent [28], given that H_2 and CO follow Henry's law. Considering this behaviour, a pressure of 40 bar was used for this study, since the rate-determining step for pressures above 35 bar is the surface reaction regime, instead of the diffusion transport regime [29]. These higher pressures thermodynamically favour a decrease in net number of moles and, as a result, produce an increase in heavier compounds.

Fig. 4 shows the variation in CO conversion versus time. The monometallic cobalt catalyst (Co10-c) records an increase in CO conversion of up to 32% after 5 h on-stream, returning a conversion value of around 26% after 25 h of reaction. The behaviour of the iron catalyst is similar to the monometallic cobalt system for short times, but after 2.5 h of reaction the conversion level drops, and the catalyst is totally deactivated after 9 h. This deactivation is attributed to the

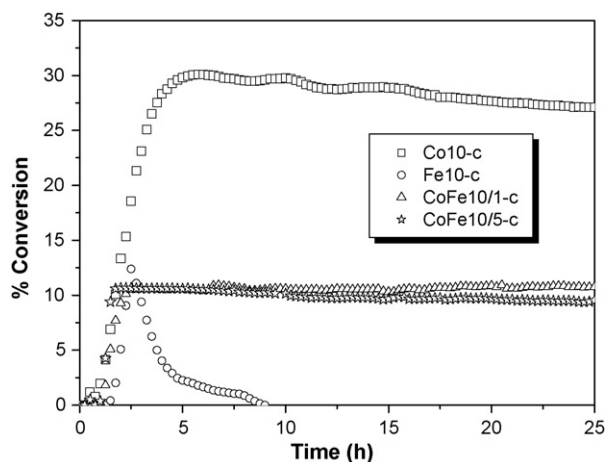


Fig. 4. CO conversion for mono- and bimetallic catalysts in a slurry reactor ($T = 503$ K, $P = 40$ bar, $H_2/CO = 2:1$).

CO disproportionation that produces carbon deposits – a reaction that is clearly more successful on iron catalysts than on their cobalt counterpart [30]. A TEM microphotograph of the monometallic iron catalyst after reaction shows the presence of helical type carbon filaments (Fig. 5). A similar deactivation, due to the formation of carbonaceous species that block the active sites, has been found on an iron catalyst by Bukur et al. [31]. They observed that this different performance could also be attributed to the lower degree of iron reduction in the slurry reactor, during Fischer–Tropsch synthesis conditions [32]. Therefore, the difference in performance of the iron catalyst in both types of reactors could be related to a further activation occurring during FTS in a fixed-bed reactor, but not in a slurry reactor. The characterization data for spent iron catalysts, found in literature [32], showed clear differences in the nature of the active phase in the catalysts employed in fixed-bed and slurry reactors, which means that differences can be related to changes in performance.

Table 3

Catalytic data for cobalt-based catalysts in a slurry reactor ($T = 503$ K, $P = 20$ bar, $H_2/CO = 2:1$.)

	Co10-c	CoFe10/1-c	CoFe10/5-c
Conversion (%)	28	11	10
C ₁	22	23	35
C ₂	3	6	6
C ₃	5	9	9
C ₄	3	5	6
C ₅₊	65	55	41
CO ₂	2	3	3
α^a	0.83	0.64	0.63
Olefinity	34.8%	48.5%	46.8%

^a Calculated for C₄–C₁₂.

CO conversion levels of bimetallic catalysts (CoFe10/1-c and CoFe10/5-c) increase to 12% for 2 h of reaction, remaining constant after this point. The addition of iron to a cobalt catalyst produces a decrease in CO conversion of around 60%.

A brief compilation of reaction results in the liquid phase reactor is presented in Table 3. As observed, the main product for all catalysts is methane, and this methane selectivity increases upon adding iron to the base cobalt catalyst. It is also observed that bimetallic catalysts produce a higher amount of light hydrocarbons and a lower amount of C₅₊ hydrocarbons than Co10-c.

Fig. 6 shows selectivity toward light hydrocarbons for Co10-c catalyst. The very low selectivity to ethane observed in Fig. 6A is due to the relative ease with which the C₂H_x fragments are re-adsorbed, and then reacting via polymerization and hydrogenolysis on the active sites [21]. Consistent with this is the higher selectivity to propene, indicating that C₂H_x fragments undergo further adsorption and reaction to yield propene on the active catalyst surface. C₂–C₆ paraffins (Fig. 6A) present similar selectivity values and are all lower than 3%, with ethane in the smallest proportion. For paraffins,

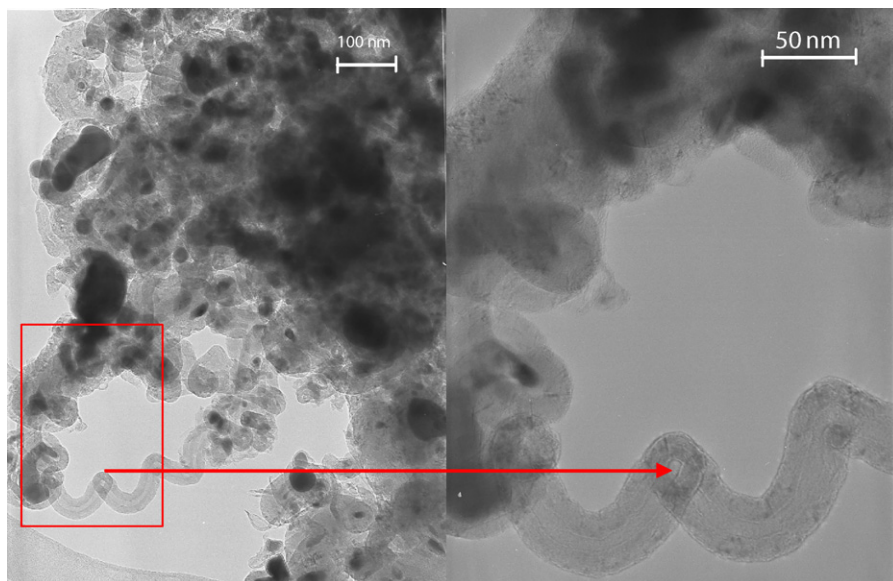


Fig. 5. TEM micrographs of Fe10-c catalyst after reaction in a slurry reactor ($T = 503$ K, $P = 40$ bar, $H_2/CO = 2:1$).

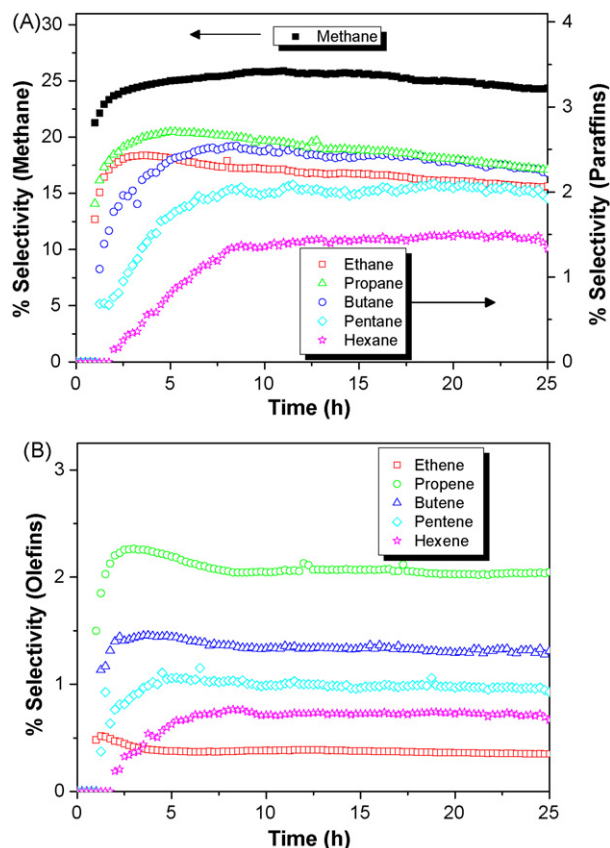


Fig. 6. Selectivities to paraffins (A) and olefins (B) for Co10-c catalysts in a slurry reactor ($T = 503$ K, $P = 40$ bar, $H_2/CO = 2:1$).

the first products that appear are C_2 – C_3 that reach maximum values at around 3–5 h, respectively, and then decrease with reaction time. This diminished is associated with the formation of C_4 – C_6 products. The time for which selectivities toward paraffins stabilize is higher than that required for olefins. This is due to the longer residence time of products, as their formation process is longer, with the following formation steps: (i) dissolution in the solvent, (ii) adsorption and (iii) hydrogenation to give paraffins.

The introduction of a small amount of iron (CoFe10/1-c) produces notable changes in selectivity. This system shows higher selectivity toward light hydrocarbons (C_1 – C_4) than the monometallic cobalt catalyst (Co10-c) (Table 3). The catalytic behaviour for paraffin formation (Fig. 7A) is very similar to that shown by the monometallic cobalt catalyst, although in this case ethane is the major product, with lower selectivity toward higher carbon number paraffins. This indicates that while C_2 species are re-adsorbed for the monometallic cobalt catalyst and a methylene group is inserted, producing an increase in chain length, C_2 species for this bimetallic catalyst are hydrogenated producing mainly ethane. Selectivities to olefins are very similar to those obtained with the monometallic cobalt catalyst, with propene being the olefin with the highest selectivity and ethane the one with lowest. The introduction of 5% wt Fe (CoFe10/5-c) leads to a catalytic behaviour similar to that observed for CoFe10/1-c catalyst,

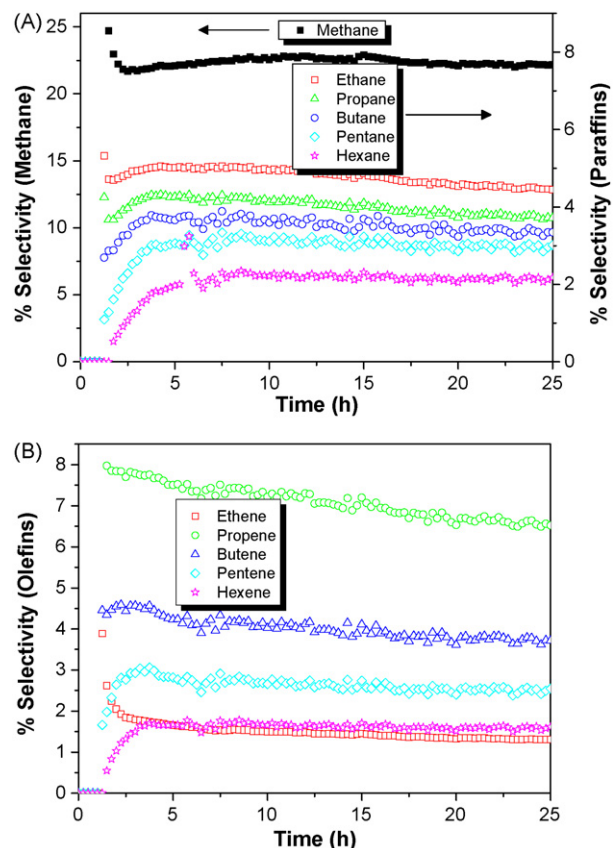


Fig. 7. Selectivities to paraffins (A) and olefins (B) for CoFe10/1-c catalysts in a slurry reactor ($T = 503$ K, $P = 40$ bar, $H_2/CO = 2:1$).

although the polymerization rate seems to decrease while hydrogenation follows the opposite trend (Figs. 8A and B).

These data thus show that iron introduction produces an increase in the selectivity to olefins by a factor of around 1.5 (Table 3). This behaviour can be determined by the combination of cobalt and iron sites in the Fe–Co alloy [33], which is suitable for the formation of C_2 – C_4 alkenes. The higher olefin selectivity found for Co–Fe bimetallic systems compared to Co monometallic catalyst is attributed to the high olefin selectivity of iron. Early studies on iron catalysts indicated that H_2 mobility was significantly restricted by blocking the low coordination edge and corner sites for dissociative adsorption on hydrogen [34]. Reduced H_2 mobility, as well as reduced hydrogen adsorption rates, could explain the decrease in the hydrogenation of alkenes to alkanes, which is related to higher olefin selectivity in bimetallic systems.

Fig. 9 presents selectivities to liquid hydrocarbons for monometallic cobalt and bimetallic catalysts, showing a symmetrical, Gaussian-type distribution of the selectivity around the solvent employed (C_{16}). Bimetallic catalysts show a decrease in selectivity to liquid products with respect to Co10-c, with this decrease being more pronounced for heavier hydrocarbons, especially those with more than 20 carbon atoms. These observations are reflected in the decrease of the chain growth probability (α , calculated for C_4 – C_{12}) (Table 3). The higher chain growth in the cobalt system may be attributed

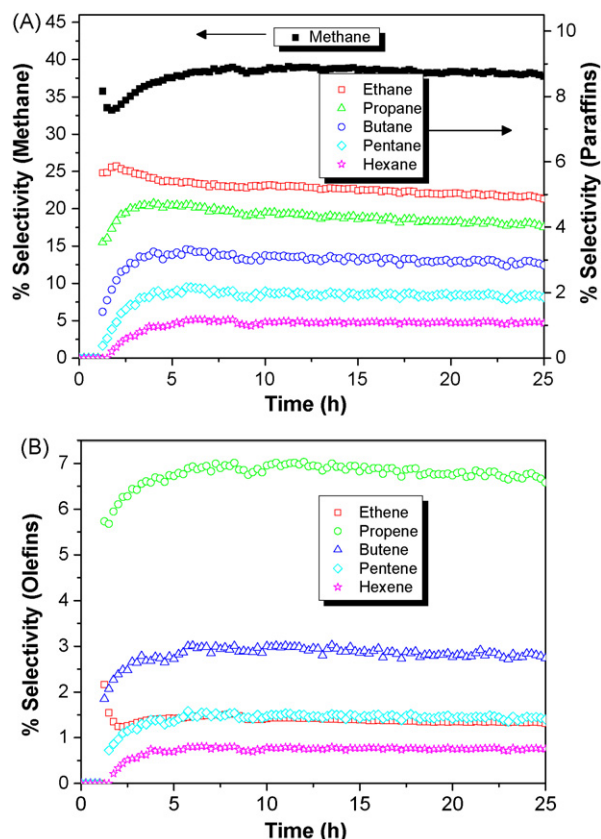


Fig. 8. Selectivities to paraffins (A) and olefins (B) for CoFe10/5-c catalysts in a slurry reactor ($T = 503$ K, $P = 40$ bar, $H_2/CO = 2:1$).

to the effective participation of olefins in the carbon-carbon chain propagation [35,36]. Preferred olefins are α -olefins (or I -alkenes) of the type $R-CH=CH_2$ wherein R is hydrogen or an alkyl group. These α -olefins can compete with carbon monoxide and heavier olefins for re-adsorption and chain initiation, and they can add directly to the growing chains. Thus, the higher α parameter value for the cobalt catalyst is related to the secondary chain growth of re-adsorbed I -alkenes, whereas the secondary chain growth is negligible for iron

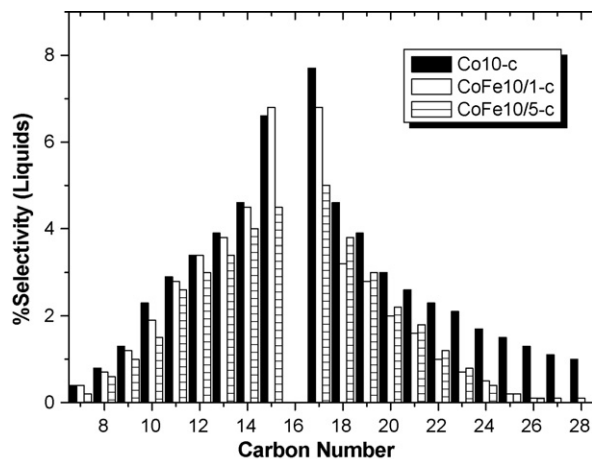


Fig. 9. Selectivities to C_7 – C_{28} hydrocarbons for cobalt-based catalysts in a slurry reactor (excluding C_{16} , solvent).

catalysts [37]. The formation of a Co–Fe alloy in bimetallic catalysts should decrease non-alloyed surface cobalt, thus decreasing the chain growth by I -alkene re-adsorption.

The absence of alcohols in reactions performed in the slurry reactor can be explained by the greater solubility of hydrogen with respect to CO in the solvent employed, as well as by the greater contact time of reactants. Both facts promote CO dissociation, and therefore decrease the possibility of a non-dissociative CO insertion [38].

4. Conclusions

The above results obtained for mono- and bimetallic cobalt and iron catalysts in FTS in both fixed-bed and slurry reactors lead to the following conclusions:

Fixed-bed reactor: (i) CO conversion is lower for the monometallic cobalt catalyst than for its iron counterpart. This behaviour is attributed to low cobalt dispersion. The increase in CO conversion upon iron incorporation is attributed to the formation of a highly dispersed cobalt–iron alloy. (ii) The cobalt catalyst favours C–C chain growth. On the other hand, the major products for iron-containing catalysts (monometallic iron catalyst and bimetallics) are light hydrocarbons, with especially high selectivity to methane for the bimetallic catalyst with 1% iron. (iii) The cobalt catalyst is less selective for alcohol formation than its iron counterpart. The cobalt–iron alloy formation in bimetallic catalysts enhances the ability for CO insertion, giving rise to an increase in selectivity toward alcohols. A high selectivity to ethanol is found for the bimetallic catalyst with 1% iron.

Slurry reactor: (i) The cobalt catalyst shows the highest CO conversion, and keeps on decreasing, but at a low rate, after 25 h of reaction. On the other hand, and after reaching 12% conversion, the iron catalyst is fully deactivated after 9 h of reaction, which is due to an accumulation of carbonaceous deposits on its surface. (ii) The cobalt catalyst gives rise to high chain growth, which decreases with the introduction of iron due to an increase in the hydrogenation reaction rate. This results in a higher methane formation rate and an enhancement of selectivity toward light hydrocarbons. (iii) The scarce production of alcohols in the slurry reactor is explained by the greater solubility of hydrogen, as well as by the longer contact time of reactants that promote CO dissociation.

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